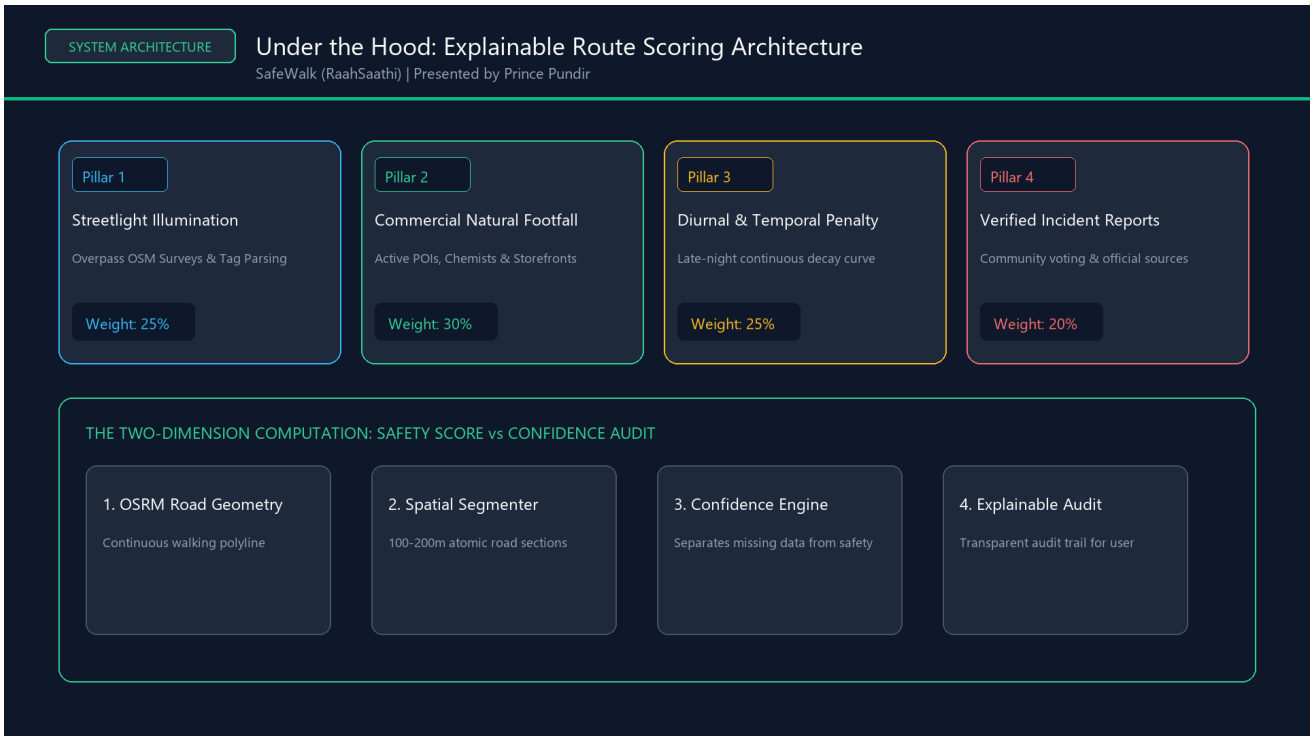
SafeWalk (RaahSaathi)

Autonomous Women's Safety Navigation & Explainable Spatial Intelligence Platform

Author / Lead Developer: Prince Pundir | **Evaluation Package:** Project Technical Dossier & Verification Guide

1. Executive Summary & Problem Dilemma

Mainstream navigation systems (Google Maps, Apple Maps) optimize solely for speed (distance divided by time). For a woman commuting alone late at night, shaving 90 seconds by routing through an unlit backroad, an abandoned underpass, or an unmonitored industrial stretch can be hazardous. **SafeWalk (RaahSaathi)** redefines pedestrian routing by prioritizing **human safety and survival over raw transit time**, backed by real road geometry, verified assistance havens, cryptographic live tracking, and explainable spatial auditing.



2. Core Architectural Modules

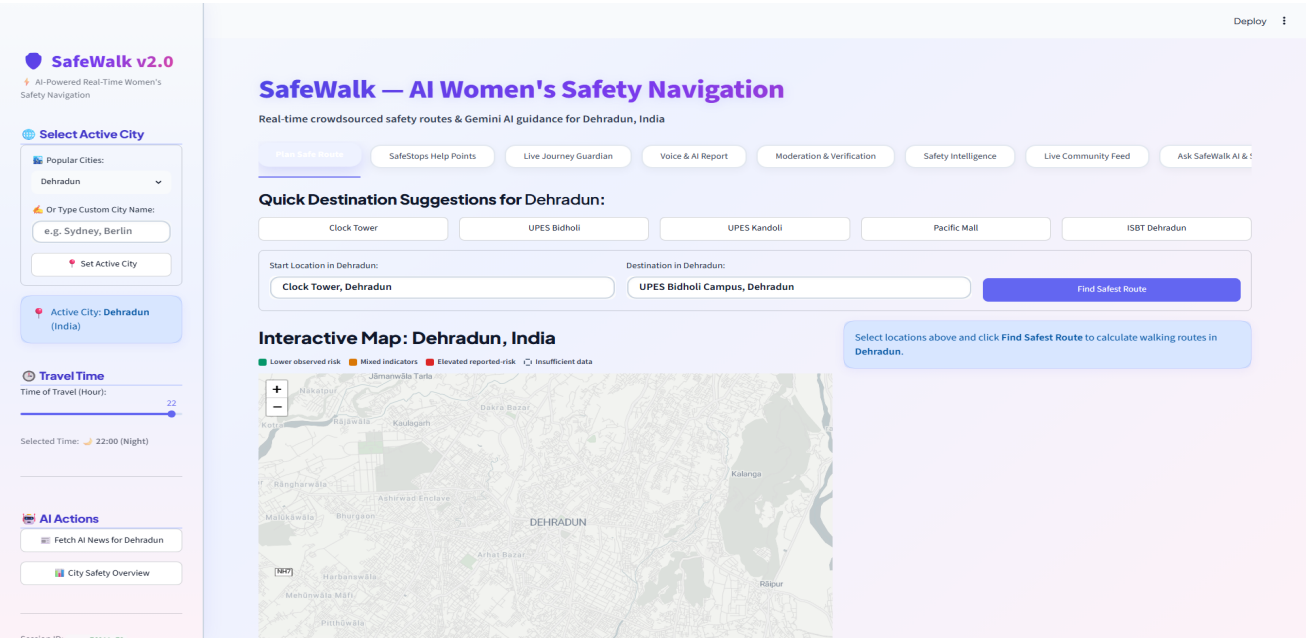
Feature Module	Technical Implementation	Safety Impact
Dual-Route Navigation	OSRM pedestrian engine + OpenStreetMap Overpass spatial queries. Compares safest path vs shortest path.	Prevents night-time routing through dark alleys and isolated industrial stretches.
Explainable Segment Intelligence	Subdivides routes into 100m–200m atomic sections. 4-Pillar continuous scoring with independent confidence grade.	Zero black-box obscurity. Unmonitored roads are strictly distinguished from safe conditions.
SafeStops Verified Havens	Curated spatial directory of 24/7 police, trauma centers, campus security desks, and late-night chemists.	Instant 1-tap dynamic rerouting to the closest safe haven with verified phone contact links.

Live Journey Guardian	32-byte cryptographic high-entropy tokens (SHA-256). Periodic automated check-in timer.	Privacy by default. Automated simulated SMS escalation with last valid GPS if check-in missed.
Multilingual AI Voice Reporting	Gemini NLP engine parsing Hindi, English, and Romanized Hinglish with human confirmation gating.	Zero mobile friction. Speak naturally in dark or emergency situations without complex forms.
PWA Offline Sync	Service Worker + IndexedDB queue with client-generated cryptographic idempotency keys.	Maintains uninterrupted navigation and local incident caching in basement transit zones.
Safety Intelligence Dashboard	1.1km generalized spatial grid cells with minimum incident thresholds for municipal and campus oversight.	Target infrastructure audits (e.g. broken streetlights) without tracking individual movements.

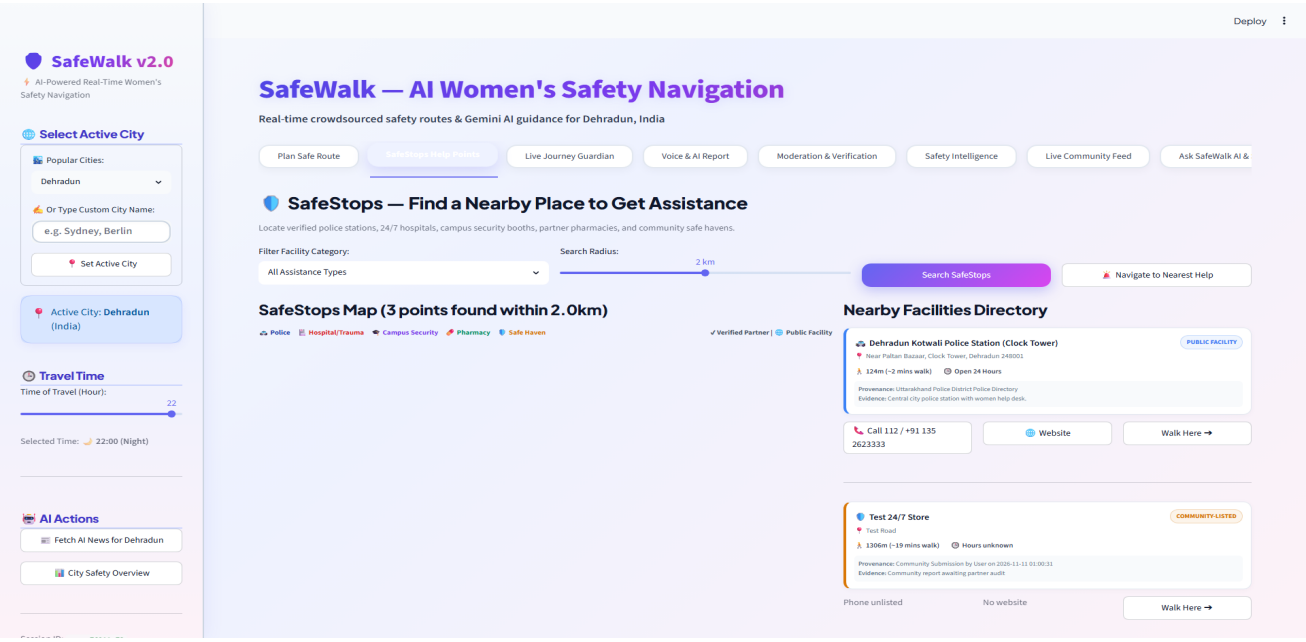
3. Production User Interface Verification

Live application screen captures taken directly from the running production instance:

Dual-Route Safe Navigation Interface



SafeStops Verified Assistance Havens Directory & Emergency Reroute



4. Technical Rigor, Automated Tests & Security

Security & Privacy by Design:
• **Zero Account Mandate:** Users do not need an account to plan routes or use emergency SafeStops.
• **Cryptographic Sharing:** Live Journey Guardian generates high-entropy 32-byte tokens. Database IDs are never exposed in URLs or payloads.
• **Data Provenance:** Open datasets (PUBLIC FACILITY) are strictly isolated from institutional partners (PARTNER-VERIFIED).
• **Aggregated Heatmaps:** Municipal dashboards enforce 1.1km cell generalization, preventing individual de-anonymization.

Automated Test Suite Verification (13/13 Passing):

All test suites run via python -m pytest tests/ with 100% pass rate:
1. test_segment_route_geometry_splits_properly — PASSED
2. test_data_confidence_scoring_independent_from_safety — PASSED
3. test_segment_analysis_explainable_breakdown — PASSED
4. test_analyze_entire_route_reasons_and_metrics — PASSED
5. test_safestops_search_and_filtering — PASSED
6. test_community_safestop_addition — PASSED
7. test_trusted_contacts_lifecycle — PASSED
8. test_journey_guardian_state_machine — PASSED
9. test_multilingual_and_hinglish_detection — PASSED
10. test_structured_incident_extraction_draft — PASSED
11. test_community_verification_and_moderation — PASSED
12. test_offline_queue_and_idempotent_sync — PASSED
13. test_safety_intelligence_aggregation — PASSED

5. Project Verification Links

Deliverable	Location / URL
GitHub Repository	https://github.com/PrincePundir123/RAAHSAATHI
Live Production App	https://raahsaathi.onrender.com (or local at http://localhost:8501)
Full Video Demo	1080p Full HD demo video with voiceover narration and subtitles
Lead Developer	Prince Pundir